

MUKESH.M

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Professional Summary

AI & ML postgraduate student with hands-on experience in Deep Learning, Computer Vision, Medical Image Analysis, and Explainable AI (XAI). Skilled in developing CNN-based segmentation and classification models using PyTorch, TensorFlow, and Scikit-learn. Experienced in Attention U-Net, semantic segmentation, Grad-CAM visualization, transfer learning, ONNX optimization, and end-to-end AI model deployment. Strong understanding of data preprocessing, feature engineering, model optimization, and healthcare AI applications.

Technical Skills

Programming	Python, SQL
Frameworks	PyTorch, TensorFlow, Scikit-learn, OpenCV
Deep Learning	CNN, Attention U-Net, Transfer Learning, Semantic Segmentation, Computer Vision
AI Concepts	Explainable AI (XAI), Feature Engineering, Data Augmentation, Model Optimization
Tools	Git, GitHub, ONNX, MySQL Workbench, Tableau, Power BI

Projects

Attention U-Net based Brain Tumor Segmentation using Explainable AI:

Developed an automated brain tumor segmentation framework using Attention U-Net on MRI scans. Achieved **98.7% Dice Score** and **97.9% IoU**, outperforming traditional segmentation approaches. Integrated attention gates and skip connections to improve tumor localization and preserve spatial features. Implemented Grad-CAM based Explainable AI visualization for interpretable clinical decision support. Technologies Used: PyTorch, OpenCV, Medical Image Processing, Deep Learning.

Lung Disease Detection and Recommendation System using Explainable AI with Metadata:

Developed a CNN-based chest X-ray classification model with metadata fusion for lung disease detection. Achieved **96% accuracy** and **0.93 F1-score** on 5,000+ labelled X-ray images. Optimized inference using ONNX for fast CPU deployment and implemented Grad-CAM visualization. Integrated a rule-based recommendation engine for symptom analysis and clinical support. NIH ChestX-ray14 and Kaggle Chest X-ray datasets.

Education

Vellore Institute of Technology (VIT)

M.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning)

2024 – 2026

CGPA: 8.22/10

KINGSTON Engineering College

Bachelor of Engineering in Mechanical Engineering
(Relevant Coursework: AutoCAD, Nx-CAD, SolidWorks, Catia and Ansys)

2019 – 2023

CGPA: 8.9/10

Certifications

- Master in Data Science & Analytics with AI – I.T Vedant, demonstrating core data analytics knowledge.
- Fundamentals of Machine Learning with AI – Hands-on ML model development and evaluation

Additional Information

- Areas of Interest: Medical Image Analysis, Deep Learning, Explainable AI, Computer Vision
- Languages Known: English, Tamil